University of Houston
Mechanical & Aerospace Engineering Department

Senior Capstone Project: Front & Rear Wing Development for a Formula SAE Car

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Blog #2

Project Constraints

Table 1: Project Constraints-For Front & Rear Wings

Constraint / Parameter	Target / Value	Justification
Minimum downforce (F_D)	≥350 [N] at 55 [mph]	Derived from tire friction modeling to expand lateral grip in autocross sections
Maximum System Mass	≤22 [lbs] (9.98 kg	Keeps total vehicle mass under 464 [lbs] target, minimizing acceleration and inertia penalties
Center of Pressure (CoP)	55%- 60% (from the front axle	Aligns aerodynamic load split with the 54% rear mechanical weight distribution to prevent high-speed imbalance.
Target Lateral Acceleration	~2 G	Target cornering capability enabled by expanded tire friction capacity
Structural Deflection	< 25 [mm] under 200 N static load	Mandatory FSAE Aerodynamics Rule; permanent deflection must remain under 5 [mm]
Dimensional Requirements	FSAE Rulebook dimensions	If these dimensional rules are not met. The vehicle will be disqualified from competition

These constraints create several competing design requirements. Increasing wing area or aerodynamic loading can increase downforce, but it also increases structural loading, mass, and drag. Increasing structural stiffness can reduce deflection but typically requires additional material and therefore increases system mass. Similarly, achieving the required 55–60% center of pressure requires careful balancing of front and rear wing loading while maintaining compliance with the dimensional envelope. The design must therefore maximize aerodynamic efficiency while simultaneously satisfying structural, mass, packaging, and vehicle-dynamics requirements.

To tackle these constraints, the team will develop a plan to test a variety of hand-picked airfoils to determine which airfoil combination, and setups will lead to the best performance. It has been decided that for the front wing design, CFD will be performed to test the following airfoils in hopes of finding a satisfactory foundation. Details can be seen below

Table 2. Airfoil Data

Category	Name	Max. Thickness		Max. Camber		C_L[A]			C_D[B]		
		%	Chord Length Location	%	Chord Length Location	α = 10°	α ~ 12.5°	α = 15°	α = 10°	α ~ 12.5°	α = 15°
Slat	AH79A	10	20.7	3.6	56.5	1.35-1.4	1.4	1.35	0.03	0.045	0.08
Slat	AH79B	10	30.9	6.4	50	1.65-1.7	1.75	1.7	0.025	0.05	0.085
Slat	NACA6409	9	29.3	6	39.6	1.4	1.45	1.5	0.025	0.04	0.065
Slat	NACA6412	12	30.1	6	39.6	1.55	1.6	1.6	0.02	0.035	0.065
Slat	E66	10.1	28.7	4.1	52.3	1.35	1.4	1.45	0.03	0.04	0.07
Slat	DAE-31	11.1	29.3	6.7	47	1.55	1.65	1.7	0.025	0.035	0.075
Slat	GOE 500	10	30	6	50	1.45	1.5	1.6	0.03	0.05	0.08
Flap	CH10	12.8	30.6	10.2	49.3	1.8	1.85	1.75	0.02	0.04	0.085
Flap	E423	12.5	23.7	9.5	41.4	1.85	1.9	1.85	0.02	0.04	0.07
A Approximate, estimated value											
B Approximate, estimated value --> Typically more chaotic distributions											

It's important to note that the Lift & Drag coefficient values are historical values obtained from separate testing at different Reynolds numbers.

The tables below include details on which airfoils will be tested and what parameters will be used for testing.

Figure 1. Angle of Attack Visual Parameter that will be tested

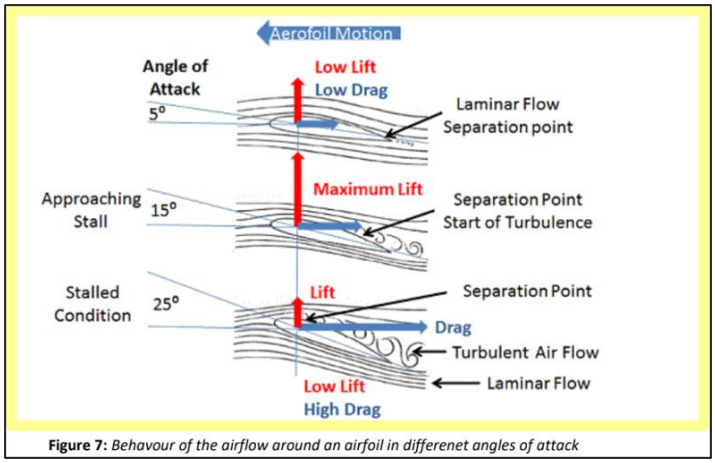


Figure 7: *Behavour of the airflow around an airfoil in differenet angles of attack*

Table 3. Parameters for Front Wing using 2 Elements

2 Element					
Case	Slat	α°	Flap	α°	Converging Gap (mm)
1	DAE-31	13-16	E423	10-13	10-13
2	DAE-31	13-16	CH10	8-11	10-13
3	AH79B	7-15	E423	10-13	10-13
4	AH79B	7-15	CH10	8-11	10-13
5	NACA6412	9-15	E423	10-13	10-13
6	NACA6412	9-15	CH10	8-11	10-13

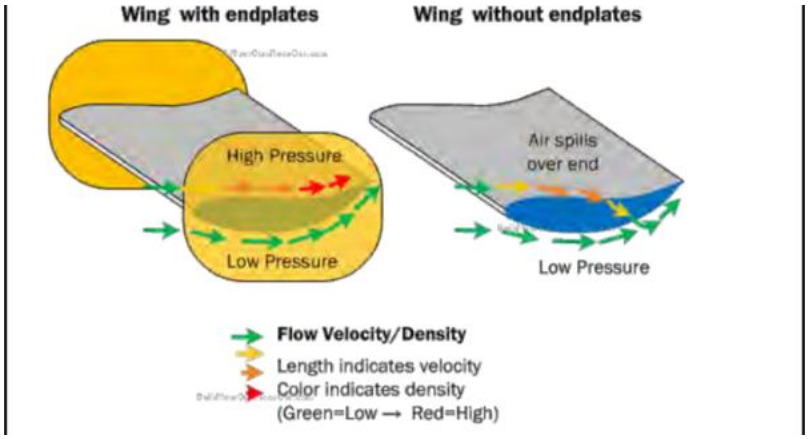
Table 4. Parameters for Front Wing using 3 Elements

3 Element								
Case	Slat	α°	Element 2	α°	Flap	α°	Converging Gap 1 (mm)	Converging Gap 2 (mm)
1	DAE-31	13-16	E423	10-13	E423	10-13	10-13	10-13
2	DAE-31	13-16	CH10	8-11	E423	10-13	10-13	10-13
3	DAE-31	7-15	CH10	8-11	CH10	8-11	10-13	10-13
4	AH79B	7-15	E423	10-13	E423	10-13	10-13	10-13
5	AH79B	7-15	CH10	8-11	E423	10-13	10-13	10-13
6	AH79B	7-15	CH10	8-11	CH10	8-11	10-13	10-13
6	NACA6412	9-15	E423	10-13	E423	10-13	10-13	10-13
6	NACA6412	9-15	CH10	8-11	E423	10-13	10-13	10-13

6	NACA6412	9-15	CH10	8-11	CH10	8-11	10-13	10-13

The main things that are to be extracted from the CFD testing are pressure distributions, velocity gradients, and lift/drag coefficients. Other important data such as boundary layer effects, stagnation points, wake, and flow separation will also be recorded for each case. Endplates (Figure 2) will also be tested to determine which geometry is the most optimal versus its weight. Due to the lack of endplate research comparted to airfoils, we except the endplate testing to be a huge learning moment for the team and hope that the CFD testing will aid us greatly. With the airfoils, we already have an idea of what results to expect, but the multi-element cases will provide us with completely new data that will entirely dictate design choices. This will greatly help the front wing team determine which airfoil + endplate combination is the most effective and will allow us to proceed with a design. The design itself will cover numbers such as span, chord length, and total wing area. For the Rear wing, the same approach will be taken; we already have a selection of airfoils to begin testing. For the sake of not making this paper extremely long, only front wing parameters will be used as an example.

Figure 2. End Plate Vs No Endplate Element



Throughout the process of completing this project, our team has encountered a few challenges that have been slowly addressed. The first one is having access to reliable computational power to be able to run CFD simulation software. This step is crucial for us to be able to design an aerodynamic front & rear wing with real world conditions, simulate an accurate boundary layer, a proper mesh, visualize accurate airflow characteristics, and obtain accurate values. Because CFD simulations require high processing power, scheduling time on shared computers and managing file transfers has demanded careful communication and planning. This has encouraged us to improve our internal workflow setting clear simulation schedules, documenting results, and maintaining version control to avoid delays. Another soft challenge has been balancing workloads and maintaining consistent progress across sub teams. We can measure success based on how efficiently our team communicates, meets internal milestones, and maintains momentum toward the final design review.

Figure 3. Visual Wing Elements

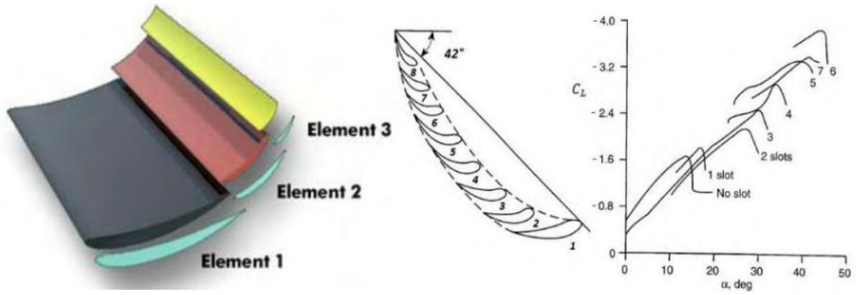


Figure 19: *Variation of lift coefficient with the number of elements*